CHAPTER 9 DIFFERENTIAL EQUATIONS

9.1 Modeling with Differential Equations

- **Differential Equation** An equation that contains an **unknown function** and some of its **derivatives**.

- **Exponential Growth Model**

> If the rate of growth of population is proportional to the population, then
>
> $$\frac{dP}{dt} = kP$$
>
> for some constant k.

- a reasonable model for a population of bacteria or animals under *ideal* conditions.
- Independent Variable = t
- Dependent Variable = P
- Show that $P(t) = Ae^{kt}$, for some constant A, is a **solution**. Verification

$\frac{dP}{dt} = Ake^{kt}$

LHS of DE $= \frac{dP}{dt}$

$= Ake^{kt}$

$= kP$ = RHS of DE (shown)

- Suppose the **initial condition** is $P(0) = 100$.

Given $P_{(0)} = 100$

$100 =$

- **Initial-Value Problem** A problem of finding a solution of the differential equation that satisfies the initial condition.

Equation : An equality relation that includes at least one unknown.

eg $x + 4 = 6$

D.E. An equality relation that includes at least one derivative of unknown function.

$$\frac{dy}{dx} + x = 4$$

Solving equation => find the unknown constant

- **Logistic Model** (*First-Order Differential Equation*)

> The logistic equation is
>
> $$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$
>
> for some constants k and M (*carrying capacity*).

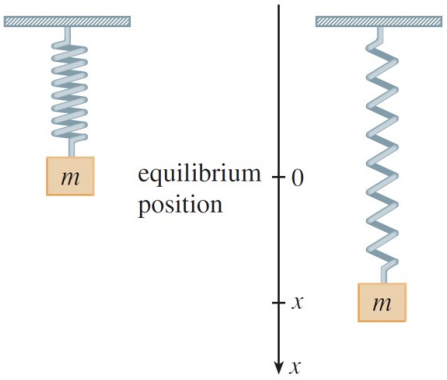
- P small:

- $P > M$:

- **Equilibrium Solutions**:

- **Mass-Spring System** (*Second-Order Differential Equation*)

> The equation of the mass-spring system is
>
> $$m\frac{d^2x}{dt^2} = -kx \qquad 2^{nd} \text{ order DE}$$



Hooke's Law:

Newton's Second Law:

Example 9.1. Verify that $y = \sin x \cos x - \cos x$ is a solution of the differential equation $y' + (\tan x)y = \cos^2 x$, on the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

$y' = \sin x(-\sin x) + \cos x(\cos x) + \sin x$

$= \cos^2 x - \sin^2 x + \sin x$

LHS of DE $= \cos^2 x - \sin^2 x + \sin x + \tan x(\sin x \cos x - \cos x)$ ($\tan x = \frac{\sin x}{\cos x}$)

$= \cos^2 x - \sin^2 x + \sin x + \sin^2 x - \sin x$

$= \cos^2 x$

= LHS of DE

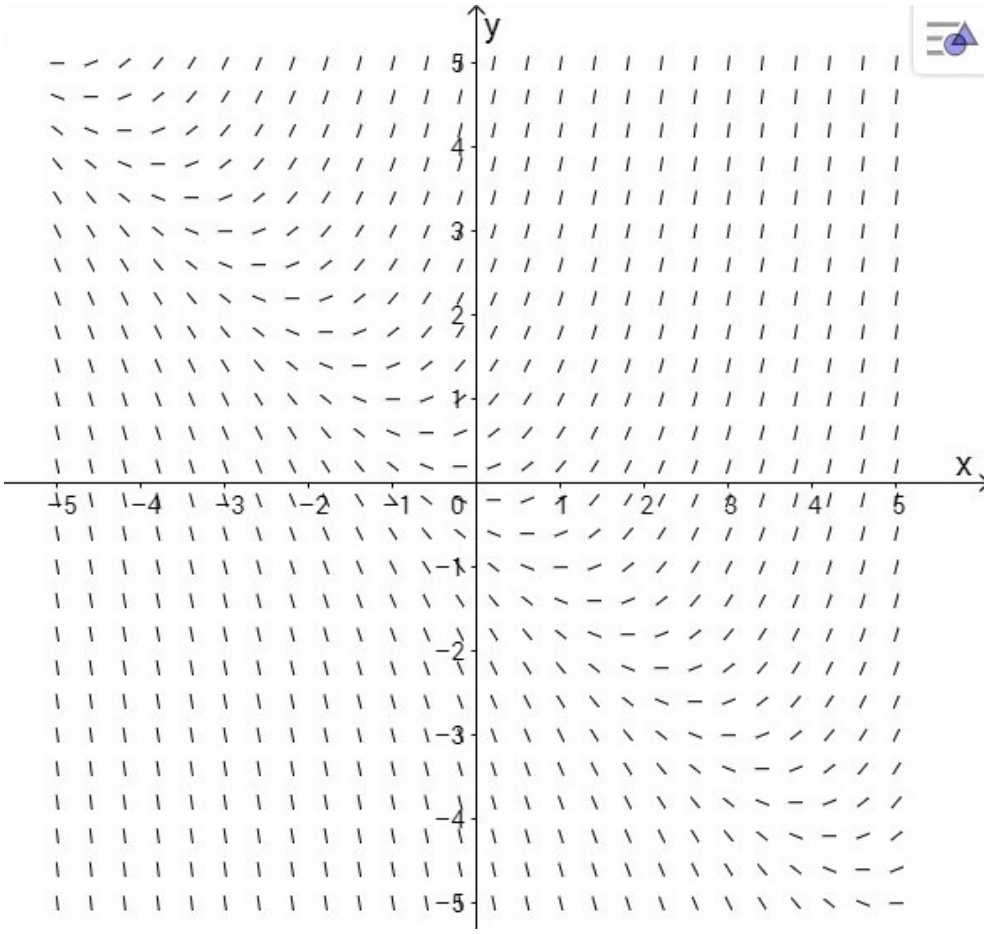
Exercises 9.1 1, 9.

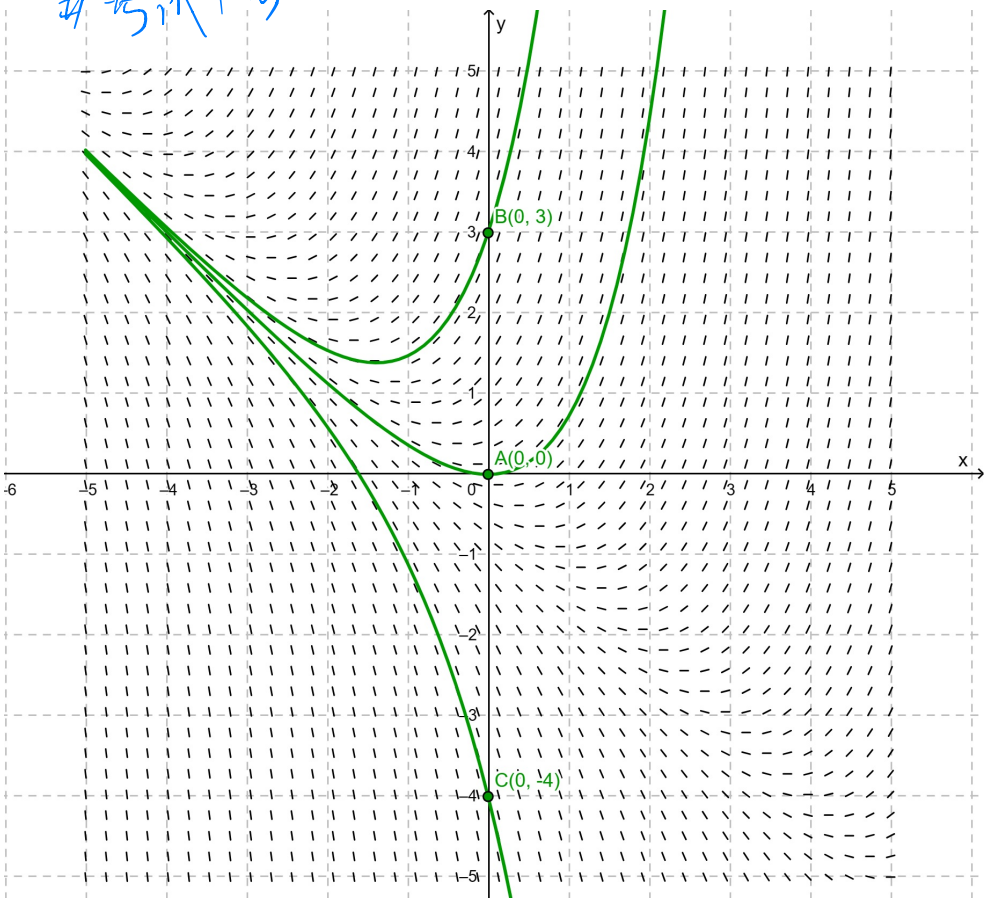
9.2 Direction Fields and Euler's Method

- Consider the first-order differential equation $y' = F(x, y)$. The differential equation says that the slope of a solution curve at a point (x, y) on the curve is $F(x, y)$. If we draw short lines of slope $F(x, y)$ at different points (x, y), the result is called a **direction field**. The line segments indicate the direction in which a solution curve is heading, so the direction field help us to visualize the general shape of the solution curves.

  Example 9.2. A direction field for the differential equation $y' = x + y$ is shown. Sketch the graphs of the solutions that satisfy the given initial conditions.

  (a) $y(0) = 0$ (b) $y(0) = 3$ (c) $y(0) = -4$




y
x
B(0, 3)
A(0,-0)
C(0, -4)

9.3 Separable Equations

$\frac{dy}{dx} = f(x) \cdot g(y)$

x 与 y 可以分开 乘在一起 or 分开

Example 9.3. Solve the differential equation $\frac{dy}{dx} = \frac{y^2}{x^2}$. Then find the particular solution that satisfies the initial condition $y(1) = 2$.

$$\frac{dy}{dx} = y^2 \cdot \left(\frac{1}{x^2}\right)$$

$$\int \frac{1}{y^2}\, dy = \int \frac{1}{x^2}\, dx$$

$$\int y^{-2}\, dy = \int x^{-2}\, dx$$

$$-y^{-1} = -x^{-1} + C$$

$$-\frac{1}{y} = -\frac{1}{x} + C$$

$$\frac{1}{y} = \frac{1}{x} - C \Rightarrow y = \frac{1}{\frac{1}{x} - C} = \frac{x}{1 - Cx}$$

explicit form

Example 9.4. Solve the differential equation

$$\frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$$

about y : move to left side

about x : move to right side

$$(2y + \cos y)\, dy = 6x^2\, dx$$

$$\int 2y + \cos y\, dy = \int 6x^2\, dx$$

$$y^2 + \sin y = 2x^3 + C$$

implicit form

Example 9.5. Solve the differential equation

$$\frac{dI}{dt} + 3I = 15$$

$\frac{dI}{dt} = (15 - 3I)(1)$

$\int \frac{1}{15-3I}\, dI = \int dt$

$-\frac{1}{3}\ln|15-3I| = t + C_1$

$\ln|15-3I| = -3t + C_2$, where $C_2 = -3C_1$

$|15-3I| = e^{-3t+C_2}$

$15-3I = Ae^{-3t}$, where $A = \pm e^{C_2}$ 不能和书中有出入

$3I = 15 - Ae^{-3t}$

$I = 5 - \left(\frac{A}{3}\right)e^{-3t}$ → change into another constant

$= 5 + Be^{-3t}$, where $B = -\frac{A}{3}$

- **Mixing Problems**

Example 9.6. A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains 0.03 kg of salt per liter of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. Let $y(t)$ be the amount of salt (in kg) after t minutes.

(a) What is $y(0)$?

$y(0) = 20$ kg

(b) Write $\frac{dy}{dt}$ in terms of y and t.

$\frac{dy}{dt} = ($ rate in of salt $) - ($ rate out of salt $)$

$= 0.03 \times 25 - \left(\frac{y}{5000}\right) 25$

$= 0.75 - \frac{y}{200}$

(c) Find the explicit solution $y(t)$.

$$\frac{dy}{dt} = 0.75 - \frac{y}{100} \quad (1)$$

$$\frac{1}{0.75 - \frac{y}{100}}\, dy = dt$$

$$\int \frac{1}{0.75 - \frac{y}{100}}\, dy = \int dt$$

$$\int \frac{100}{150 - y}\, dy = \int dt$$

$$-100 \ln|150 - y| = t + C$$

$$|150 - y| = e^{-\frac{t}{100} - \frac{C}{100}}$$

$$150 - y = Ae^{-\frac{t}{100}}$$

where $A = \pm e^{-\frac{C}{100}}$

$$y = 150 - Ae^{-\frac{t}{100}}$$

$\because y(0) = 20$

$20 = 150 - A \Rightarrow A = 130$

$$y(t) = 150 - 130e^{-\frac{t}{100}}$$

If not let $A = \pm e^{-\frac{C}{100}}$

$y = 150 \pm Ae^{-\frac{t}{100}}$

when $y(0) = 20 \Rightarrow$

$20 = 150 + A$

$A = -130$

$20 = 150 - A$

$A = 130$

$t = 30 \text{ min}$

(d) How much salt remains in the tank after half an hour?

$$y(30) = 150 - 130e^{-\frac{30}{100}}$$

$$= 38.1079 \text{ kg}$$

Exercises 9.3 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 29, 39, 43, 45.

9.4 Models for Population Growth

- **Exponential Model**

If $P(t)$ is the value of a quantity at time t and if the rate of change of P with respect to t is proportional to its size $P(t)$, then

$$\frac{dP}{dt} = kP$$

The solution of this equation is

$$P(t) = P_0 e^{kt}$$

where $P_0 = P(0)$ is the initial condition.

$\frac{dP}{dt} = kP$ P和t是变量，k是constant

$\frac{1}{P}dP = k\,dt$?

$\int \frac{1}{P}\,dP = \int k\,dt$

$\ln|P| = kt + C$

$|P| = e^{kt+C}$

$P = Ae^{kt}$, where $A = \pm e^{C}$

let $P(0) = P_0$

$P(0) = A$

$P(t) = P_0 e^{kt}$

- **Logistic Model**

The logistic equation is

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$ carrying capacity

for some constants k and M. The solution of this model is

$$P(t) = \frac{M}{1 + Ae^{-kt}}, \quad A = \frac{M - P_0}{P_0}$$

where $P_0 = P(0)$ is the initial condition.

$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$

$\int \frac{1}{P\left(1 - \frac{P}{M}\right)} dP = \int k\, dt$

$\int \frac{M}{P(M-P)} dP = \int k\, dt$ partial fraction

$\int \frac{A}{P} - \frac{B}{M-P} dP = \int k\, dt$

$\ln|P| - \ln|M-P| = kt + C$

$\ln\left|\frac{P}{M-P}\right| = kt + C$

$\left|\frac{P}{M-P}\right| = e^{kt+C}$

$\frac{P}{M-P} = \pm e^{kt+C}$

$\frac{P}{M-P} = Ae^{kt}$, where $A = \pm e^{C}$

$\frac{M}{P} =$

$\frac{P}{M} = \frac{1}{Be^{kt}+1}$

$P = \frac{M}{Be^{kt}+1}$

let $P(0) = P_0$

$P(0) = \frac{M}{B+1}$

$B = \frac{M}{P_0} - 1$

$= \frac{M - P_0}{P_0}$

Example 9.7. Given the initial value problem

$$\frac{dP}{dt} = 0.00008P(1000 - P), \quad P(0) = 100$$

(a) This is a logistic model. What is k and M?

(b) What is the solution $P(t)$?

(c) Find the populatioin size $P(40)$.

(d) Find the populatioin size $P(80)$.

(e) When does the population reach 900?

Exercises 9.4 1, 5, 9, 21(c), 22(a).

9.5 Linear Equations

- A first-order differential equation is called **linear** if it can be put into the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where P and Q are continuous functions on a given interval.

Example 9.8. $y' + \frac{1}{x}y = 2$.

Let integrating factor, $I = e^{\int P(x)\,dx}$

$= e^{\int \frac{1}{x}\,dx}$

$= e^{\ln|x|}$

$= |x|$, we choose $I = x$

$I(y' + \frac{1}{x}y) = 2I$

$(Iy)' = 2I$, $I = x$

$(xy)' = 2x$

$xy = \int 2x\,dx$

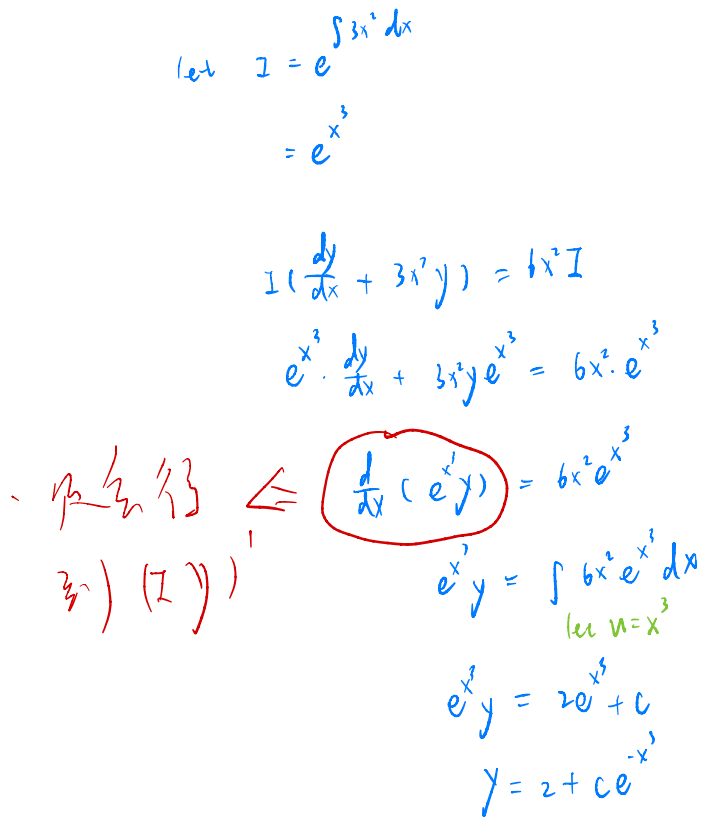
$xy = x^2 + C$

$y = x + \frac{C}{x}$

- **Integrating Factor** $I(x) = e^{\int P(x)dx}$

$$I(x)y = \int I(x)Q(x)dx$$

Example 9.9. Solve the differential equation $\frac{dy}{dx} + 3x^2y = 6x^2$.



Example 9.10. Find the solution of the initial-value problem

$$x^2y' + xy = 1, \quad x > 0, \quad y(1) = 2$$

$y' + \frac{1}{x}y = \frac{1}{x}$

Example 9.11. Solve $y' + 2xy = 1$.

$V = 100$L

Example 9.12. A tank contains 100 L of water. A solution with a salt concentration of 0.4 kg/L is added at a rate of 5 L/min. The solution is kept mixed and is drained from the tank at a rate of 3 L/min. Let $y(t)$ be the amount of salt (in kg) after t minutes.

(a) What is $y(0)$?

$y(0) = 0$

(b) Show that

$$\frac{dy}{dt} = 2 - \frac{3y}{100 + 2t}$$

$\frac{dy}{dt}$ = (rate in of salt) − (rate out of salt)

$= 0.4 \times 5 - \frac{y}{100+2t} \times 3$

in − out

5 − 3

(c) Solve the differential equation in (b) with the initial condition in (a).

$$\frac{dy}{dt} = 2 - \frac{3y}{100+2t}$$

$$\frac{dy}{dt} + \frac{3}{100+2t}y = 2$$

let $I = e^{\int P(x)\,dx}$

$= e^{\int \frac{3}{100+2t}\,dt}$

$= e^{\frac{3}{2}\ln|100+2t|}$

$= e^{\ln|100+2t|^{\frac{3}{2}}}$

$= (100+2t)^{\frac{3}{2}}$

$(100+2t)^{\frac{3}{2}}\left(\frac{dy}{dt} + \frac{3}{100+2t}y\right) = 2(100+2t)^{\frac{3}{2}}$

$\frac{d}{dt}\left[(100+2t)^{\frac{3}{2}}y\right] = 2(100+2t)^{\frac{3}{2}}$

$(100+2t)^{\frac{3}{2}}y = \int 2(100+2t)^{\frac{3}{2}}\,dt$

$(100+2t)^{\frac{3}{2}}y = \frac{2}{5}(100+2t)^{\frac{5}{2}} + C$

$y = \frac{2}{5}(100+2t) + C(100+2t)^{-\frac{3}{2}}$

let $y(0) = 0$

$0 = \frac{2}{5}\times 100 + C(100)^{-\frac{3}{2}}$

$C = 40000$

$y = 40 + \frac{4}{5}t + 40000(100+2t)^{-\frac{3}{2}}$

Integrating Factor $I(x) = e^{\int P(x)dx}$

$$I(x)y = \int I(x)Q(x)dx$$

(d) Find the concentration after 20 minutes.

$y(20) \approx 31.8527$ kg

$\text{concentration} = \frac{31.8527}{100+2(20)}$

$= 0.2737$ kg/L

Exercises 9.5 5, 7, 9, 11, 13, 15, 17, 19, 33, 35.